

MINYA, Egypt

WMO: 623870

Lat: 28.083N

Lon: 30.733E

Elev: 122

StdP: 14.63

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.1	42.1	30.0	24.2	53.8	32.4	27.0	54.6	20.7	63.0	18.1	61.2	2.5	340	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.8	104.3	71.5	101.6	71.5	99.4	71.2	76.2	93.6	75.1	92.7	74.1	91.9	7.2	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.8	118.2	82.8	70.2	111.6	82.5	68.8	106.2	82.4	39.8	93.3	38.7	92.3	37.7	91.9	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	16.6	14.8	DB	36.6	111.0	2.6	3.3	34.7	113.4	33.1	115.4	31.7	117.2	29.8	119.6
			WB	34.3	79.8	2.7	2.7	32.3	81.7	30.7	83.2	29.2	84.7	27.2	86.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.8	55.5	58.7	64.5	72.1	80.3	84.8	85.8	86.2	83.5	76.2	66.3	58.3
	DBStd	12.00	3.56	4.96	5.76	6.12	5.36	3.96	2.87	2.69	3.33	4.71	4.66	3.86
	HDD50	3	2	1	0	0	0	0	0	0	0	0	0	0
	HDD65	814	295	188	78	6	0	0	0	0	0	0	36	211
	CDD50	8309	173	243	451	664	939	1043	1110	1122	1004	812	490	258
	CDD65	3646	1	11	64	221	474	593	645	657	554	347	76	3
	CDH74	46454	26	156	789	2622	6075	8128	8969	9051	6785	3219	590	44
	CDH80	25524	2	39	279	1234	3411	4822	5258	5282	3710	1351	131	5
Wind	WSAvg	6.1	4.7	5.1	6.3	7.2	7.7	8.0	6.0	5.8	7.2	6.1	5.0	4.5
Precipitation	PrecAvg	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecMax	0.7	0.2	0.7	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.4	0.2	0.0
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.2	0.0	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.2	85.2	94.4	102.8	108.3	108.6	105.6	104.4	104.2	98.3	88.9	79.0
		MCWB	60.7	64.8	67.6	69.9	69.3	71.6	73.7	74.0	72.0	71.0	67.0	62.5
	2%	DB	72.2	78.3	87.1	97.0	102.1	104.1	101.6	101.5	100.4	93.6	83.4	73.8
		MCWB	57.9	60.1	64.1	67.4	67.7	70.7	73.3	73.6	71.0	69.7	66.2	59.8
	5%	DB	68.9	74.1	82.0	91.7	98.4	100.3	99.3	99.2	97.1	90.4	80.0	70.9
		MCWB	56.1	58.1	61.9	65.3	67.3	70.3	72.7	73.0	70.9	69.1	64.4	58.4
	10%	DB	66.4	70.6	77.7	87.0	94.7	97.2	97.1	97.2	94.4	87.2	77.1	68.5
		MCWB	54.7	56.4	59.9	63.6	66.4	69.6	72.0	72.5	70.3	68.2	63.1	57.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.5	66.0	69.5	72.2	71.8	75.0	77.5	78.2	75.6	74.5	70.3	65.1
		MCDB	74.7	83.4	89.8	99.1	98.5	97.9	96.2	93.7	92.6	90.1	82.3	74.7
	2%	WB	59.1	61.4	65.6	68.7	69.9	73.5	75.9	76.5	74.2	72.3	68.1	61.5
		MCDB	69.8	75.1	83.5	92.8	95.8	96.1	94.3	92.9	91.8	87.5	80.0	71.4
	5%	WB	57.0	59.0	62.9	66.4	68.5	72.1	74.7	75.5	73.0	70.7	66.0	59.4
		MCDB	67.1	71.9	79.8	88.6	93.4	94.4	93.0	92.0	90.6	85.5	77.2	69.3
	10%	WB	55.4	57.1	60.6	64.4	67.4	70.9	73.6	74.5	71.9	69.3	64.2	57.9
		MCDB	65.4	69.2	76.1	84.8	91.3	92.8	91.9	91.1	88.9	83.6	74.8	67.3
Mean Daily Temperature Range	5% DB	MDBR	23.3	23.9	24.9	27.1	26.7	25.2	23.7	22.8	22.7	21.5	22.2	22.2
		MCDBR	26.5	28.5	30.7	34.7	32.3	29.5	26.2	24.9	26.9	25.4	24.8	24.4
		MCWBR	15.1	14.0	13.9	13.1	9.8	8.4	8.1	7.7	7.7	8.6	11.4	12.9
	5% WB	MCDBR	25.6	26.7	29.3	32.4	29.1	26.3	24.4	22.9	23.8	22.3	22.8	23.2
	MCWBR	15.3	13.9	14.2	13.5	9.8	8.4	8.3	7.9	7.9	8.7	11.4	13.1	
Clear-Sky Solar Irradiance	taub	0.403	0.438	0.484	0.561	0.574	0.494	0.481	0.473	0.481	0.481	0.419	0.401	
	taud	2.040	1.938	1.820	1.670	1.673	1.874	1.928	1.955	1.923	1.905	2.034	2.057	
	Ebn at Noon	259	260	256	240	237	256	259	260	253	243	251	252	
	Edh at Noon	46	55	66	79	79	64	61	59	59	56	46	43	
All-Sky Solar Radiation	RadAvg	1309	1600	2005	2296	2484	2628	2560	2392	2126	1733	1396	1218	
	RadStd	33	62	58	49	57	28	24	24	42	32	33	26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.03	+1.35	N/A	N/A	N/A	+1.21	N/A	N/A	N/A	N/A
Regional (2 neighbors)		+0.79	+0.83	+1.22	N/A	+0.90	+1.53	N/A	N/A	N/A	N/A

Nomenclature: See separate page